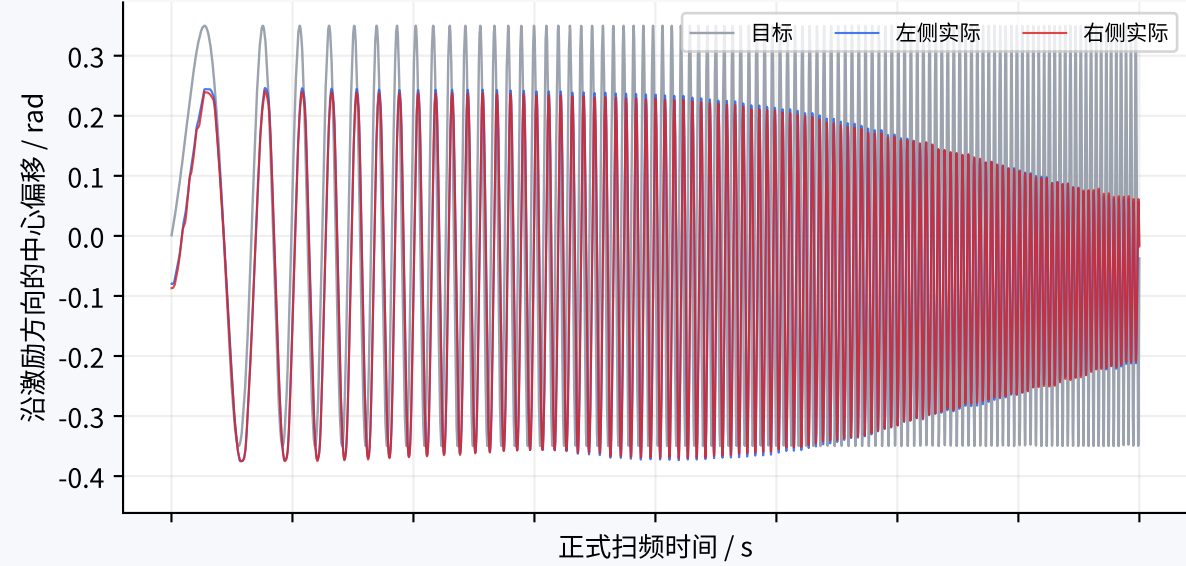
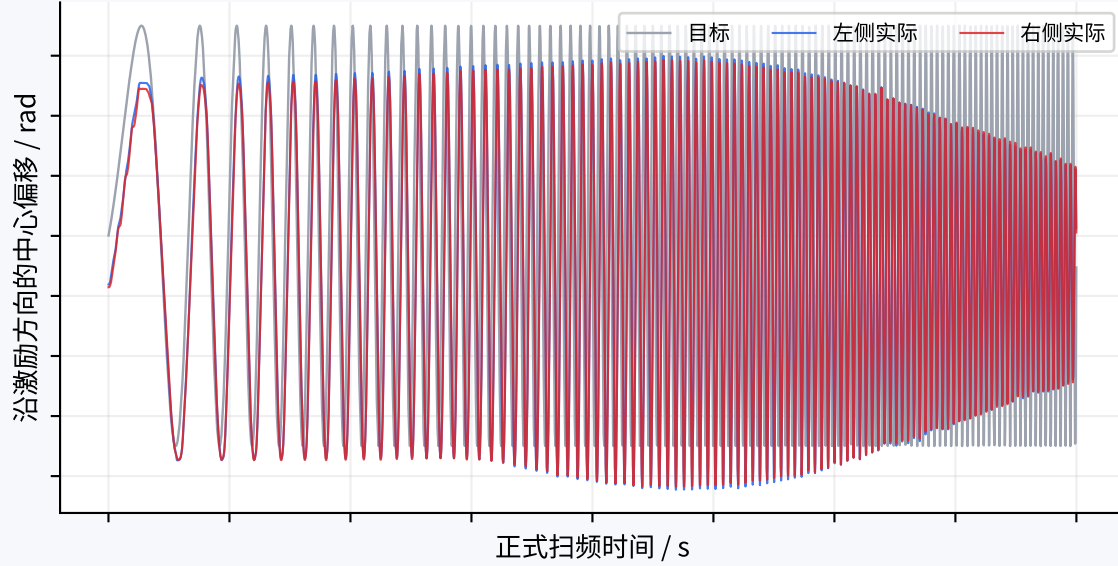


PE03 三组 PD 扫频：目标与左右实际位置  
仅绘制正式 40 秒；左右按 phase\_sign 镜像对齐，保留跟踪偏差

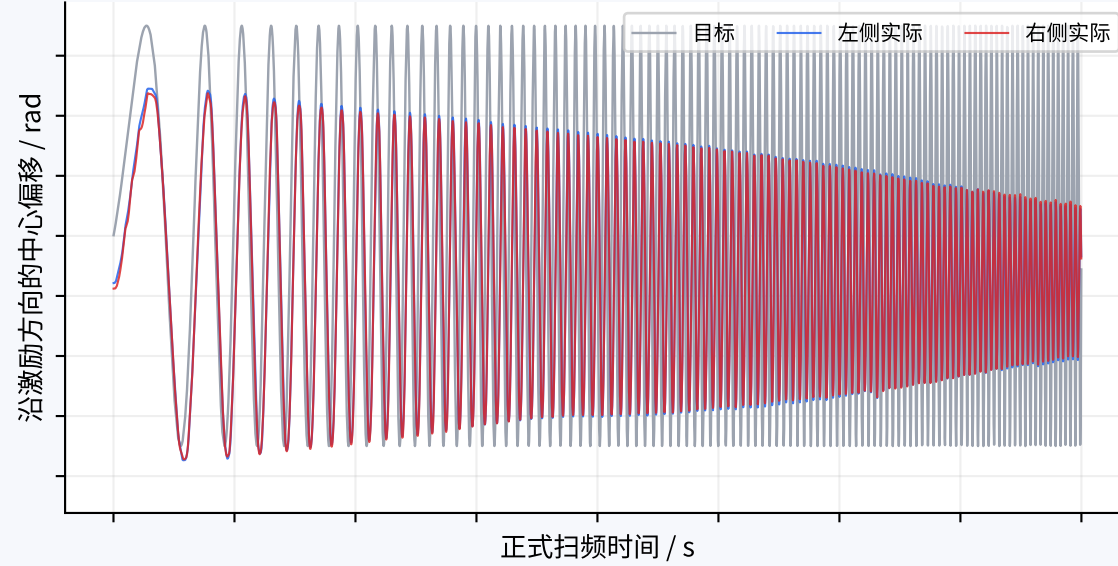
髋 hip · A 基准  
Kp=4.3 Kd=0.34 A=0.35 rad



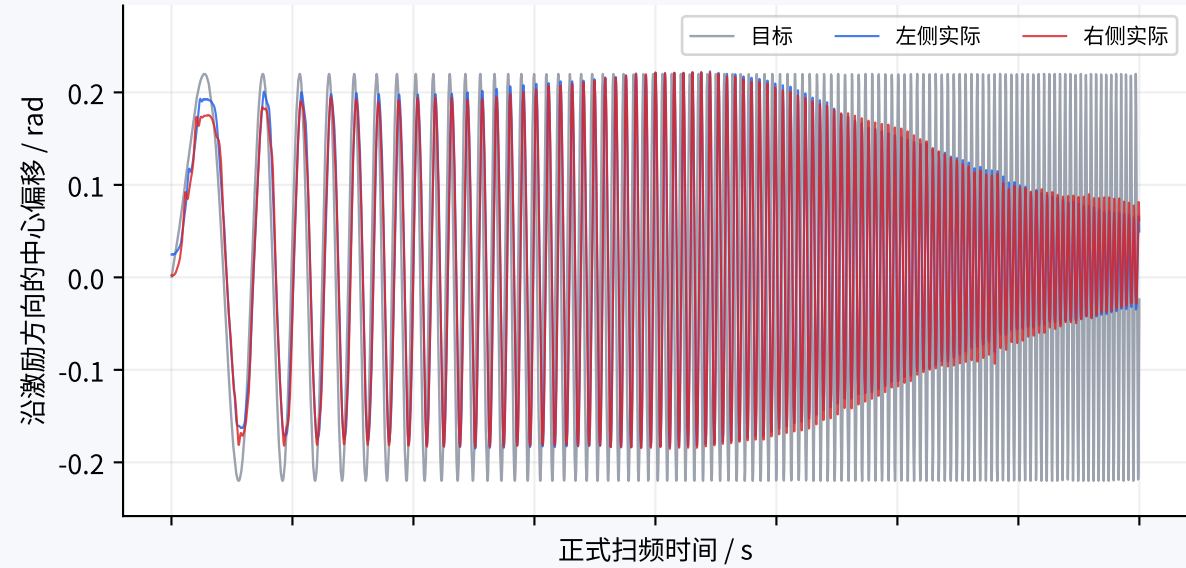
髋 hip · B 提高 Kp  
Kp=5.2 Kd=0.34 A=0.35 rad



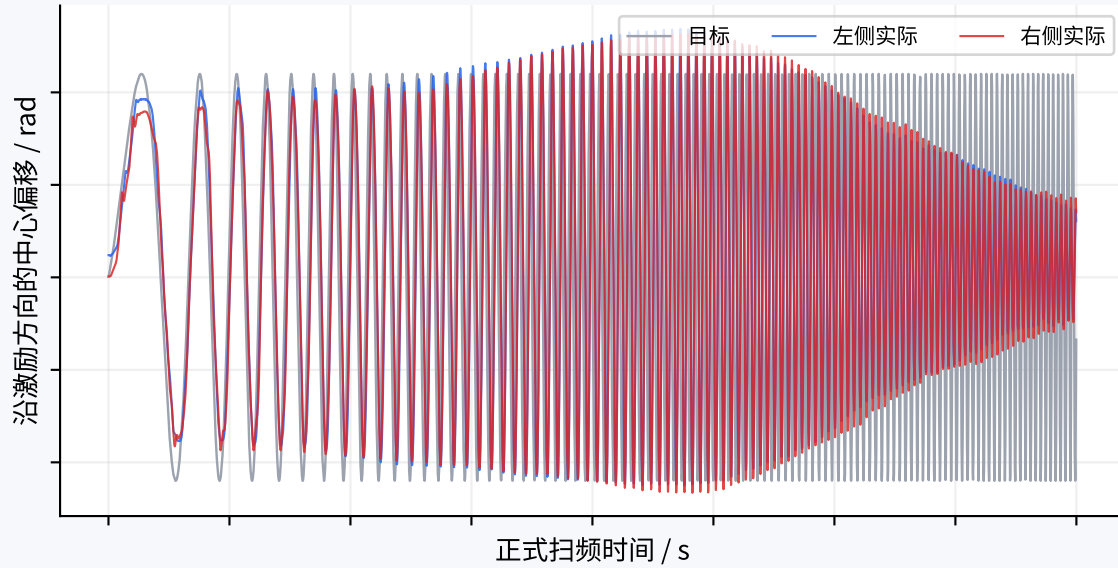
髋 hip · C 提高 Kd  
Kp=4.3 Kd=0.42 A=0.35 rad



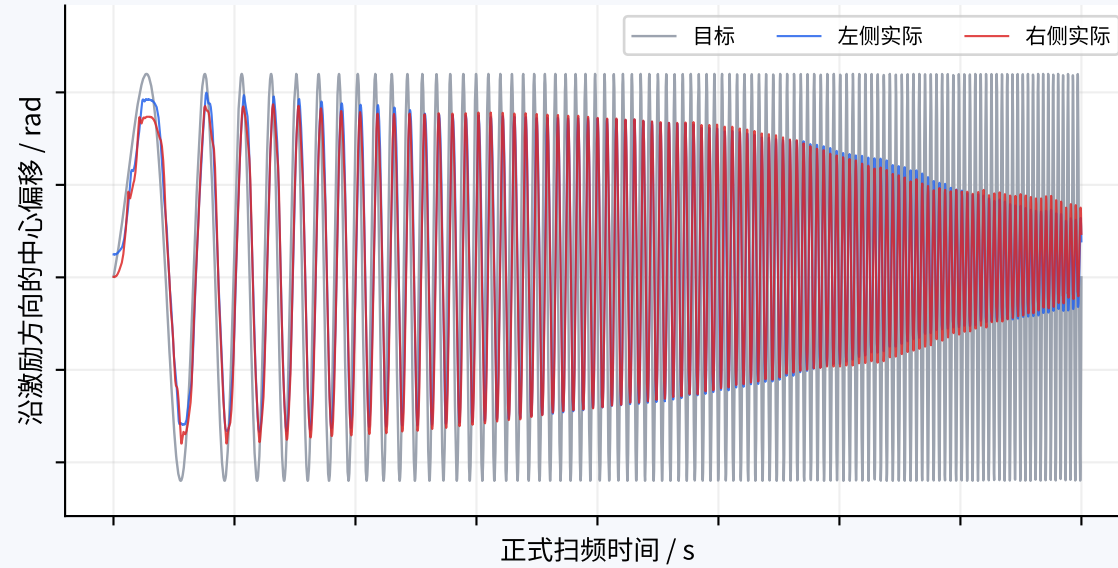
大腿 thigh · A 基准  
Kp=4.3 Kd=0.34 A=0.22 rad



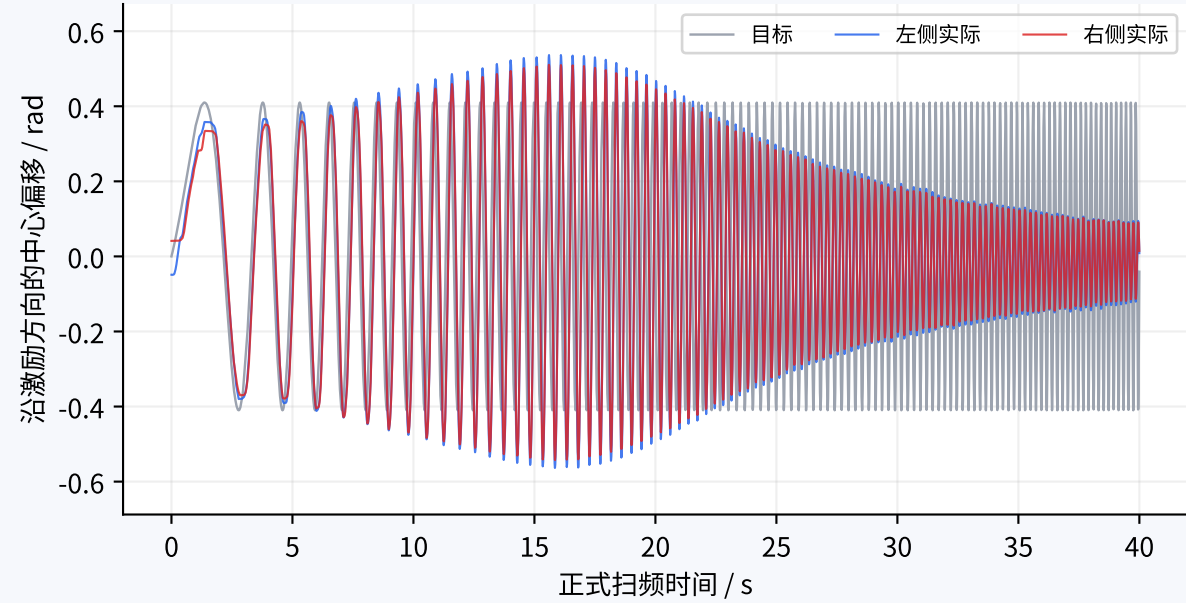
大腿 thigh · B 提高 Kp  
Kp=5.2 Kd=0.34 A=0.22 rad



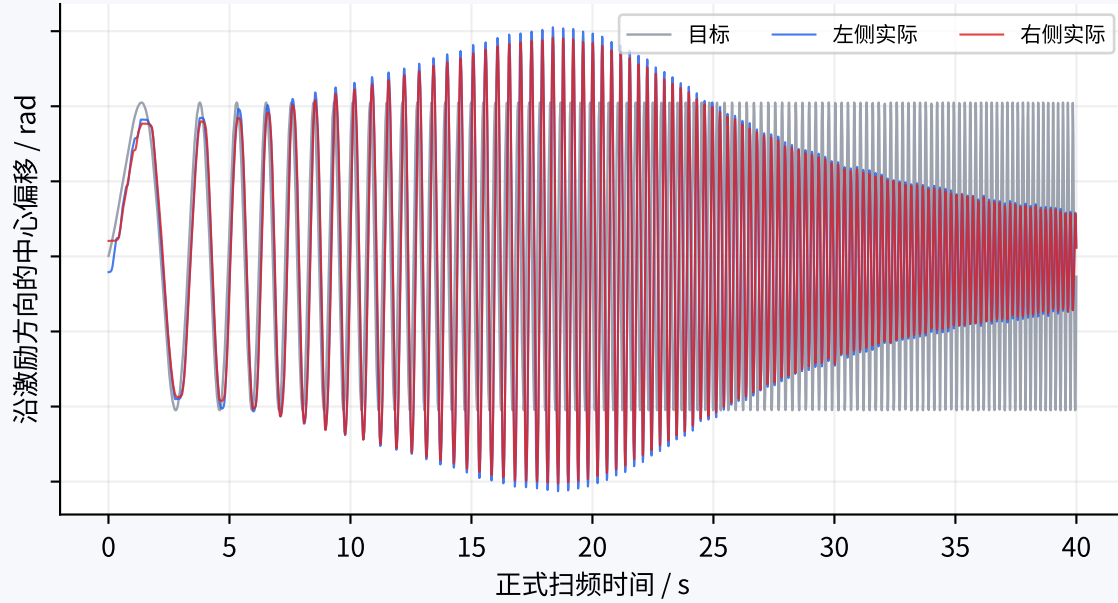
大腿 thigh · C 提高 Kd  
Kp=4.3 Kd=0.42 A=0.22 rad



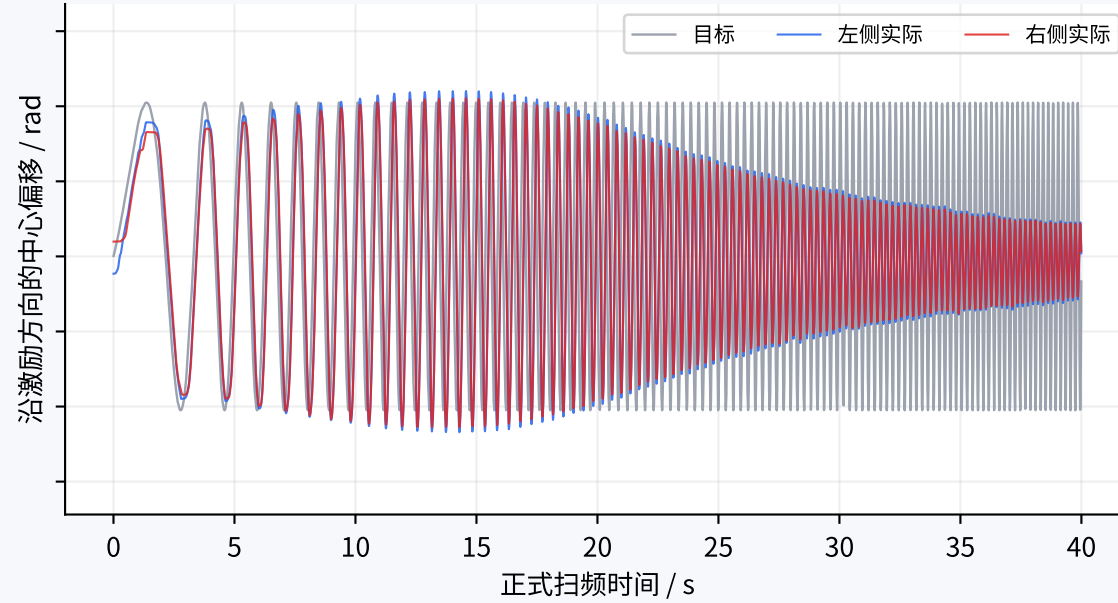
小腿 calf · A 基准  
Kp=4.9 Kd=0.24 A=0.41 rad



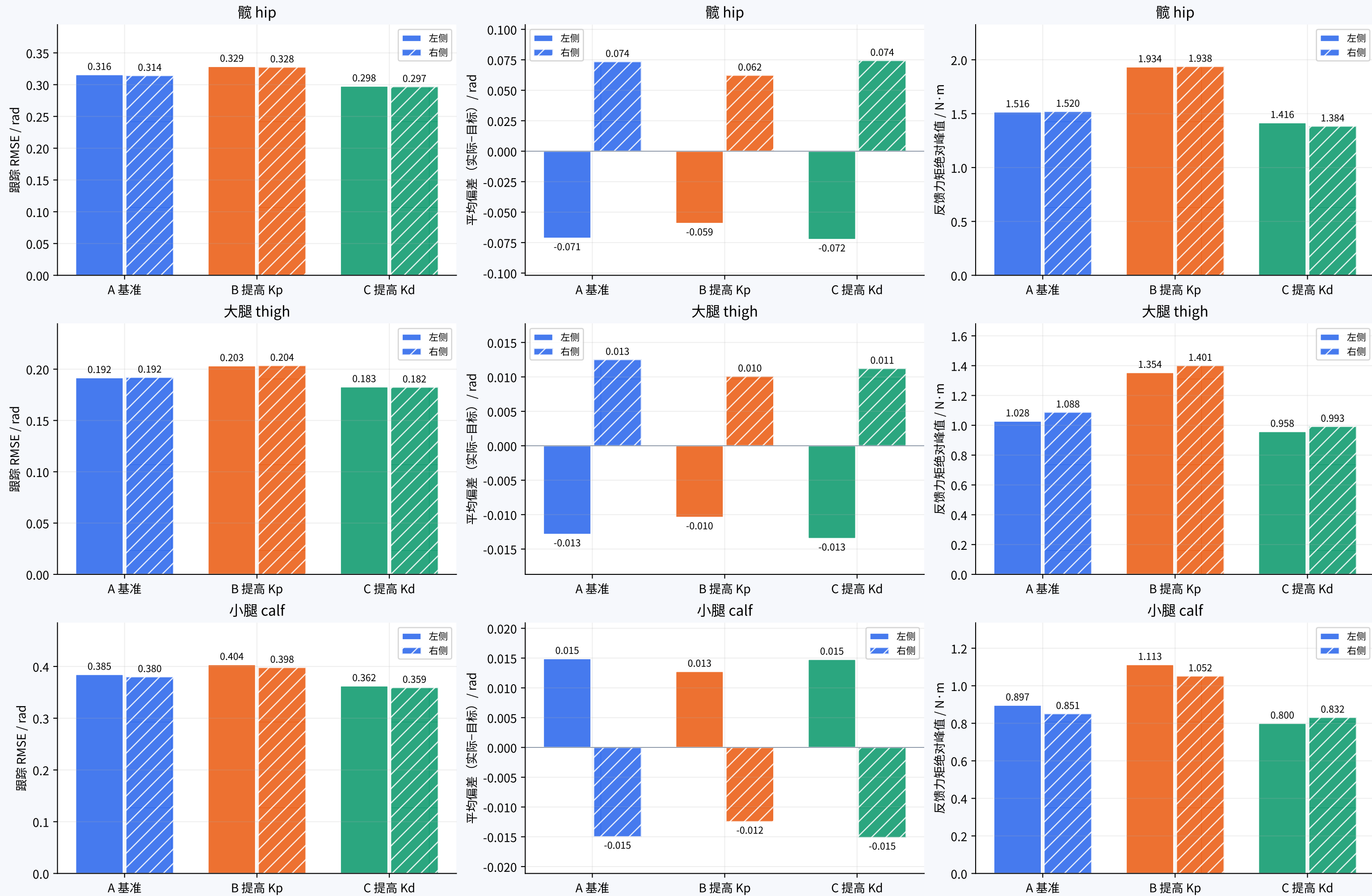
小腿 calf · B 提高 Kp  
Kp=5.9 Kd=0.24 A=0.41 rad



小腿 calf · C 提高 Kd  
Kp=4.9 Kd=0.3 A=0.41 rad

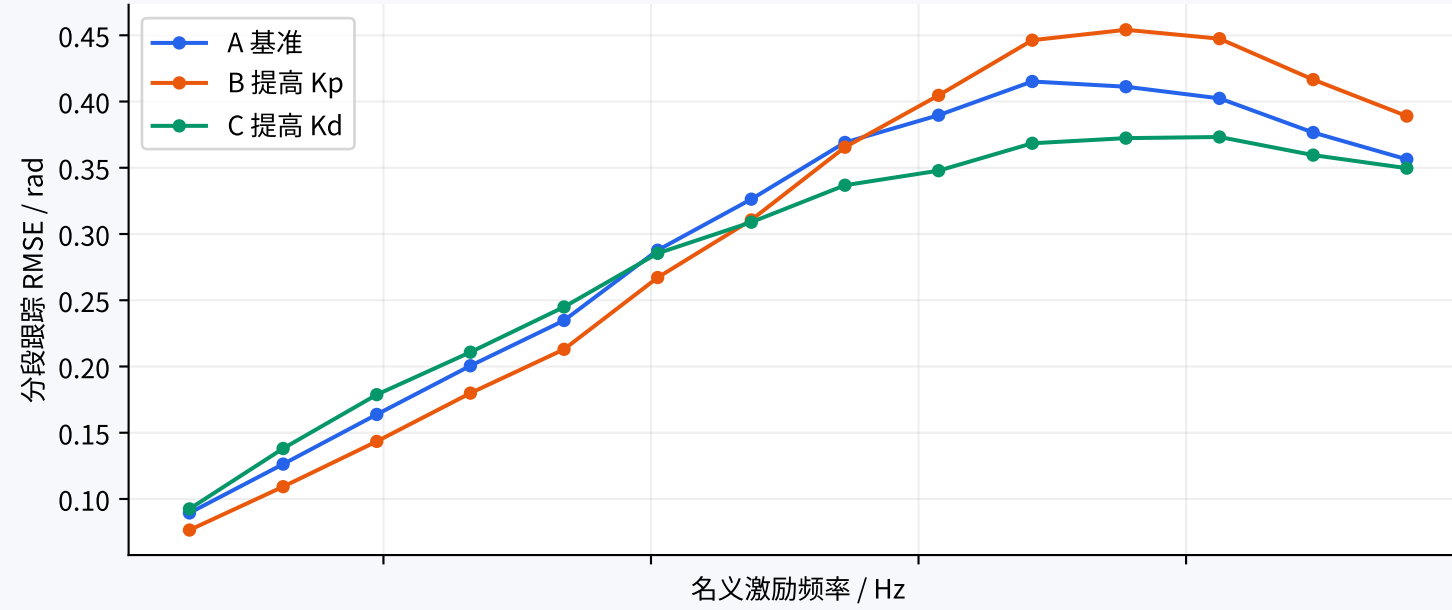


三组 PD 的记录值对比（正式扫频段）  
RMSE / 偏差按时间加权；未补偿命令应用延迟；力矩是设备反馈值

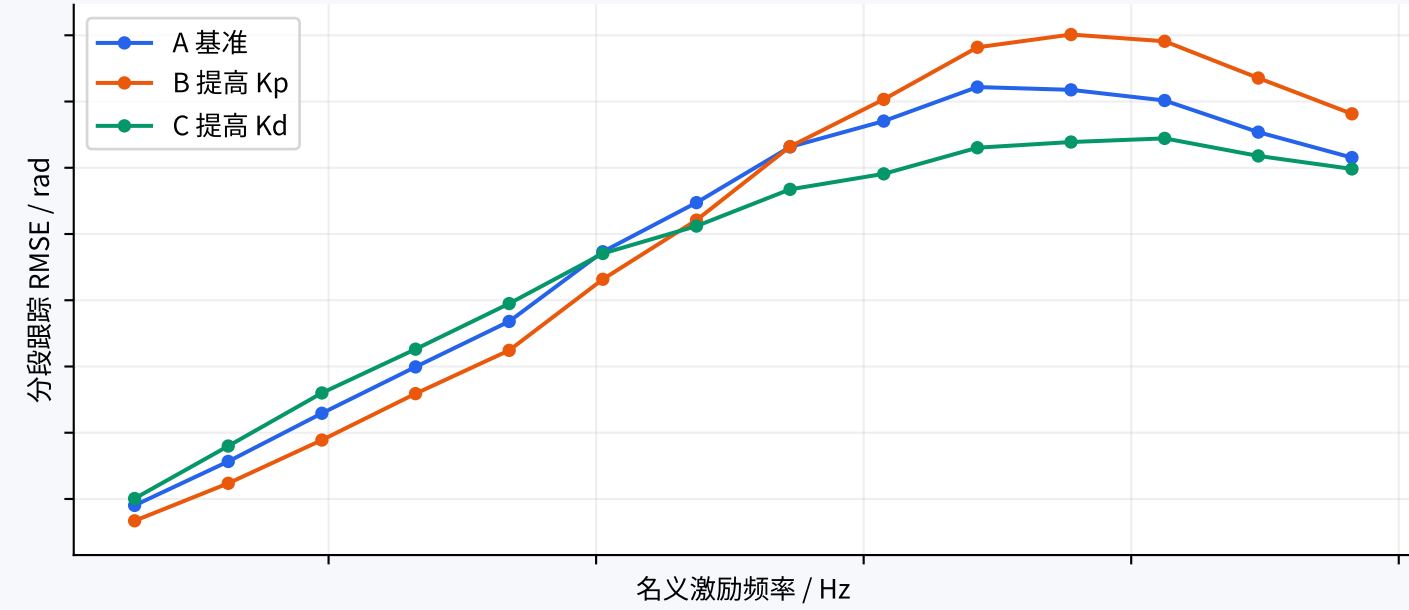


跟踪误差随扫频进程变化  
横轴由记录目标拟合出的线性 chirp 时间映射；这是分段误差，不是频率响应/带宽辨识

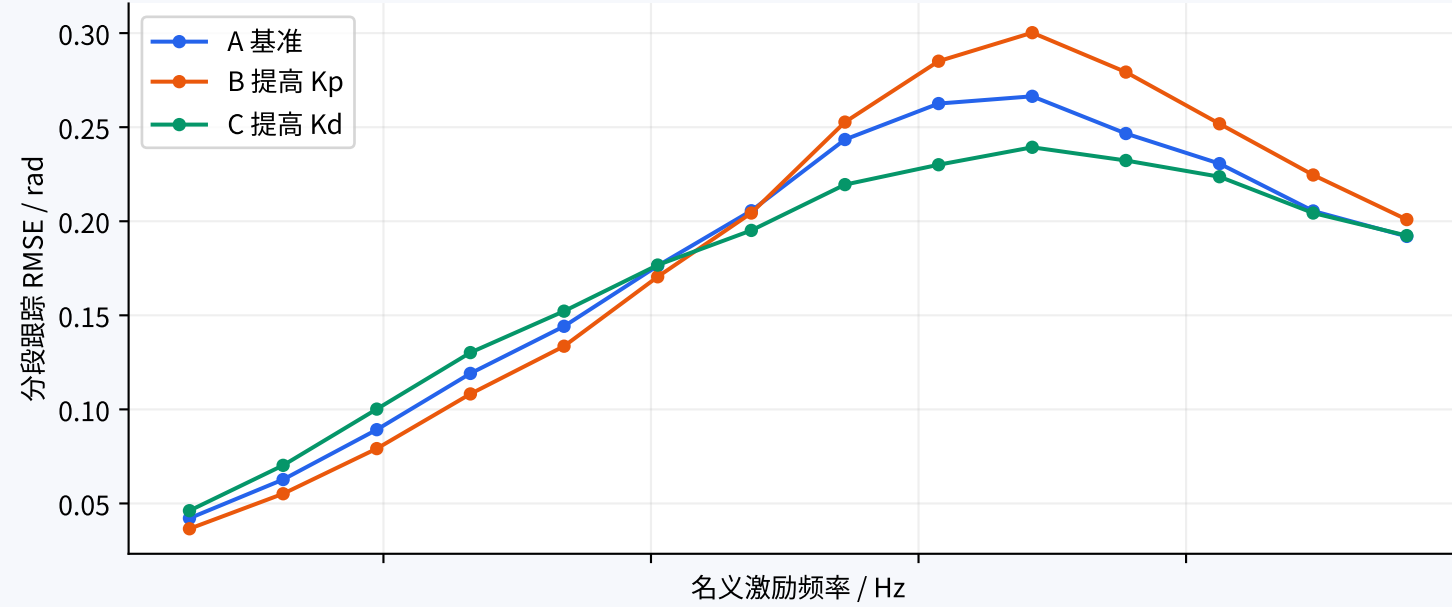
髋 hip · 左侧



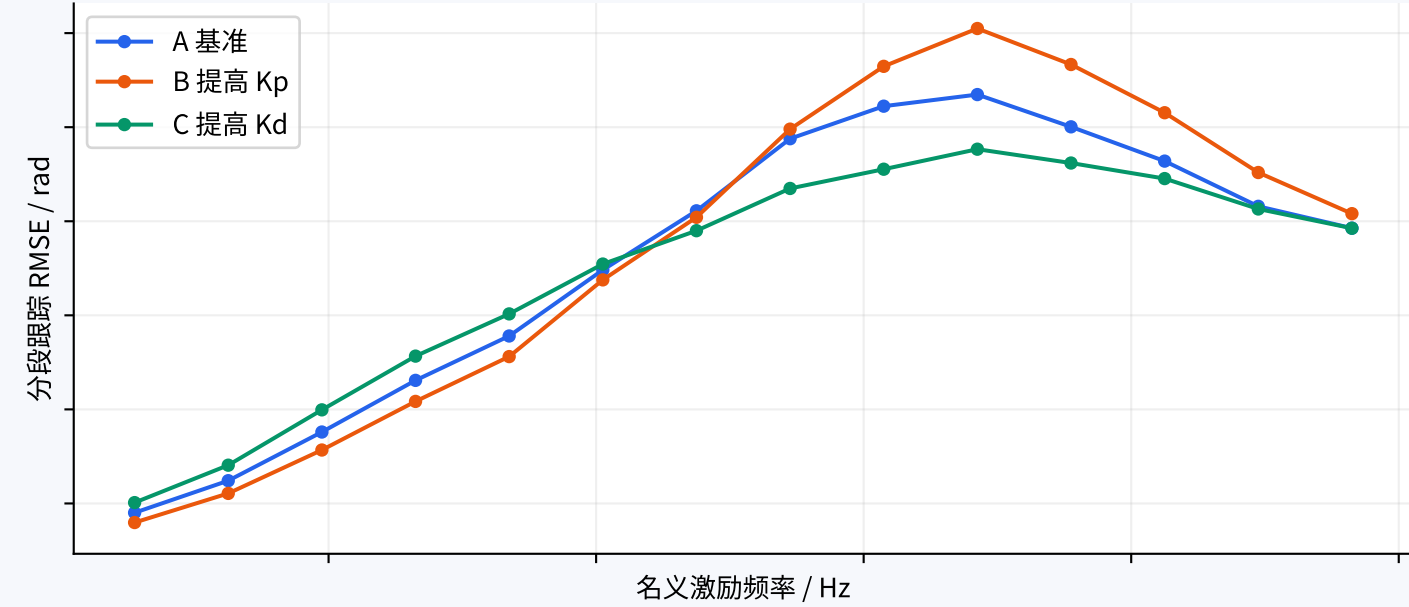
髋 hip · 右侧



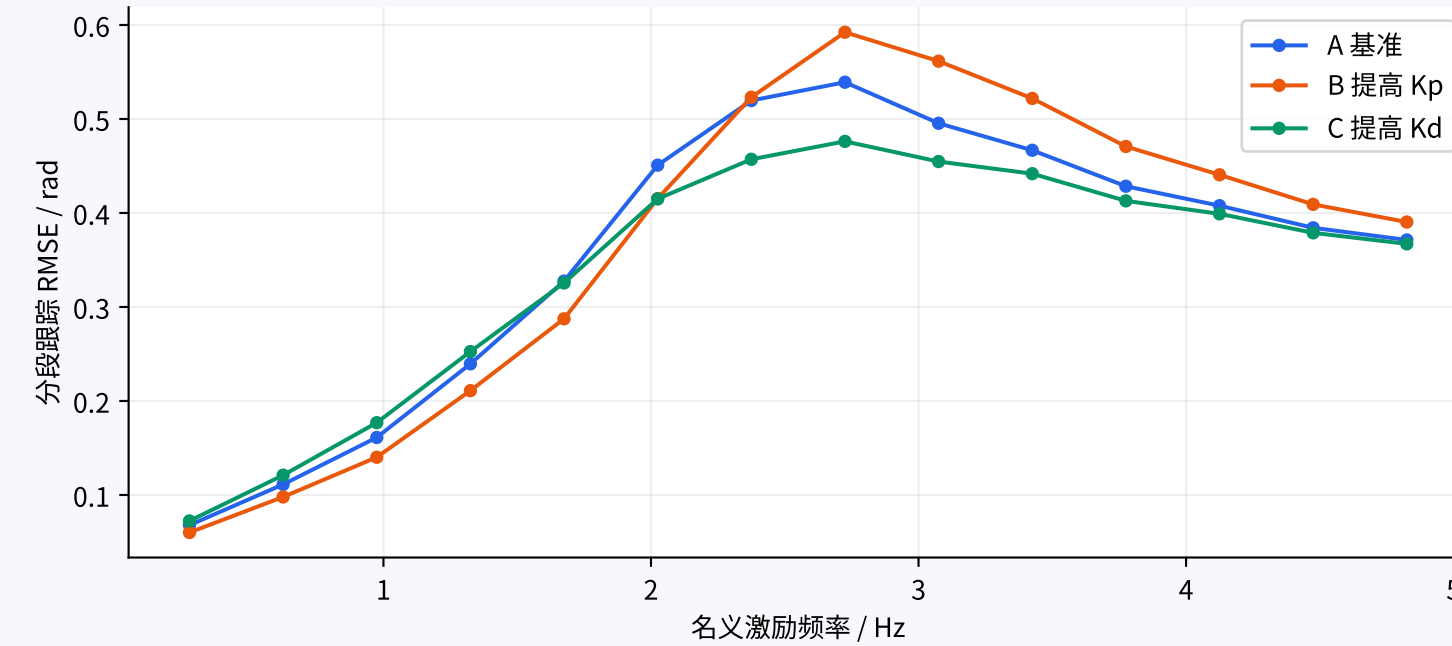
大腿 thigh · 左侧



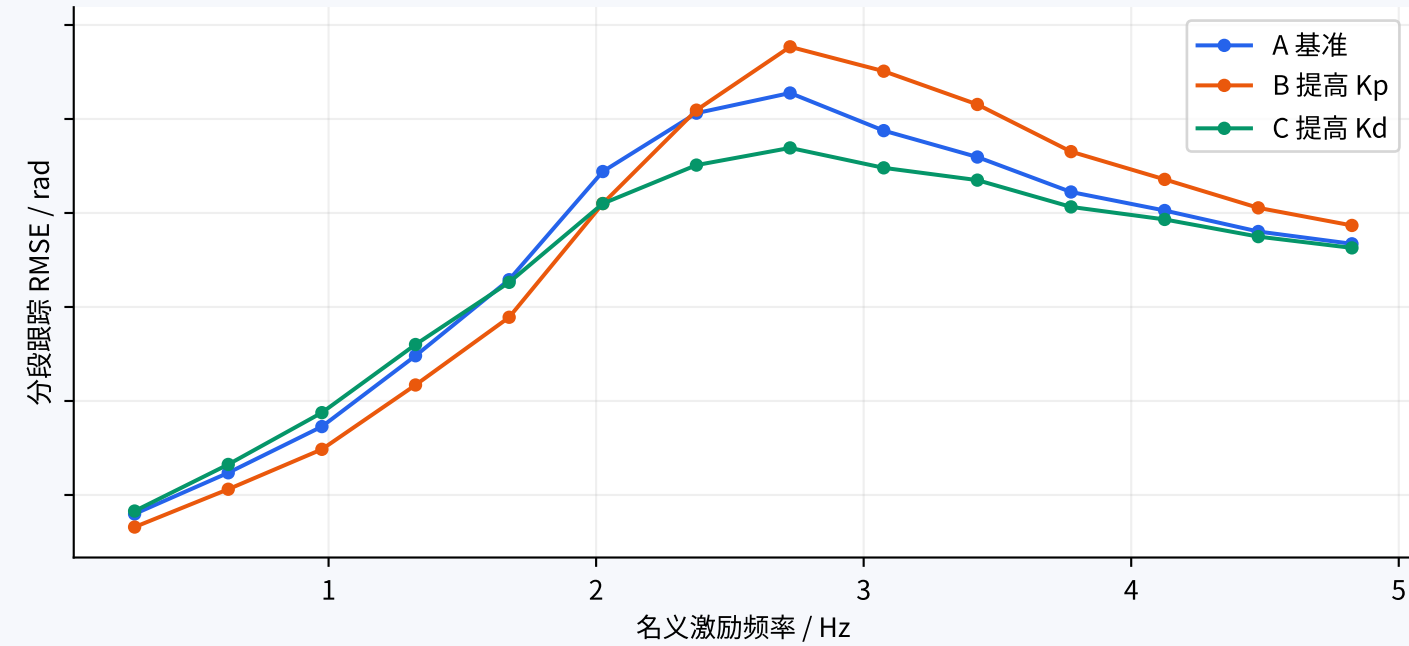
大腿 thigh · 右侧



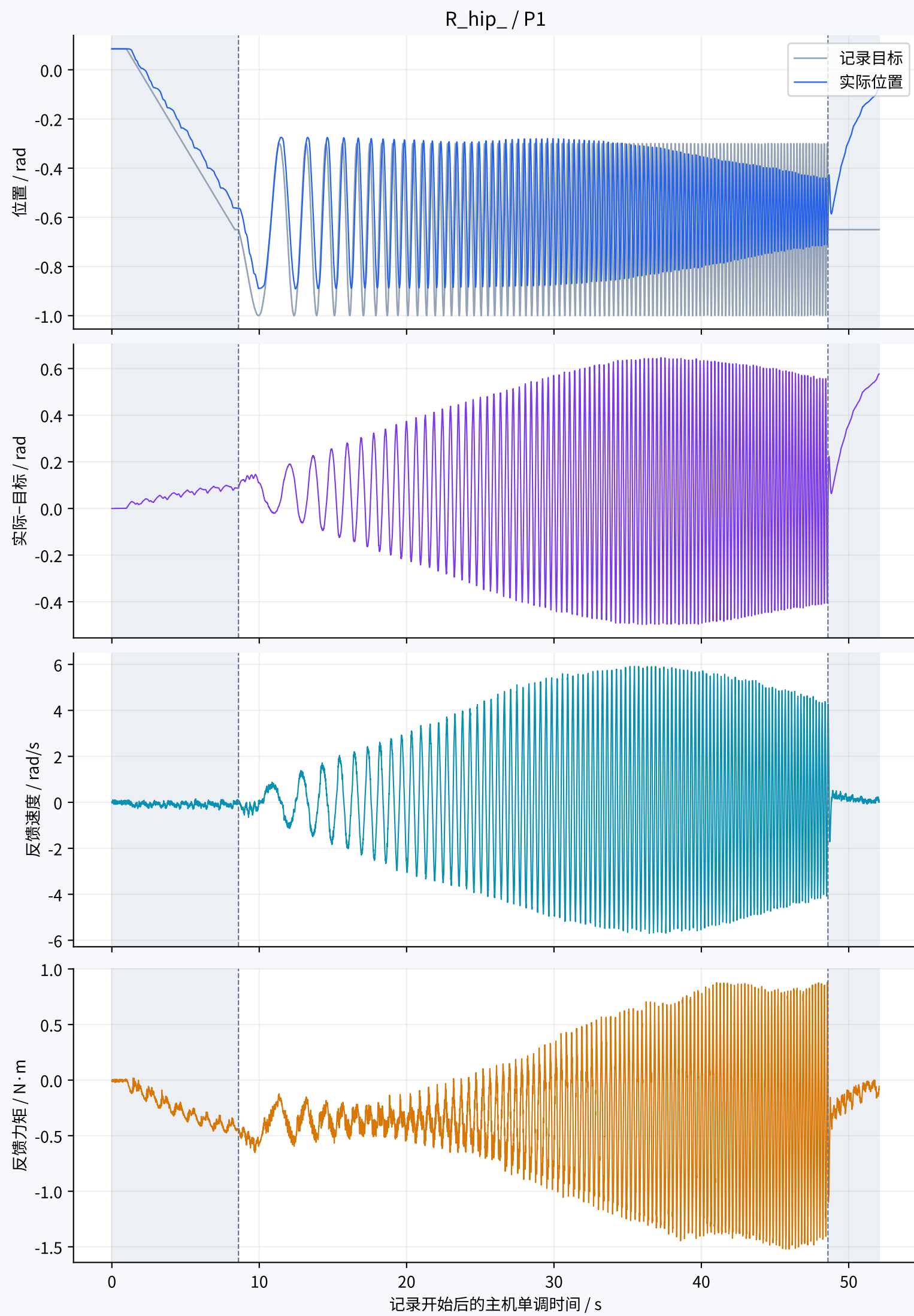
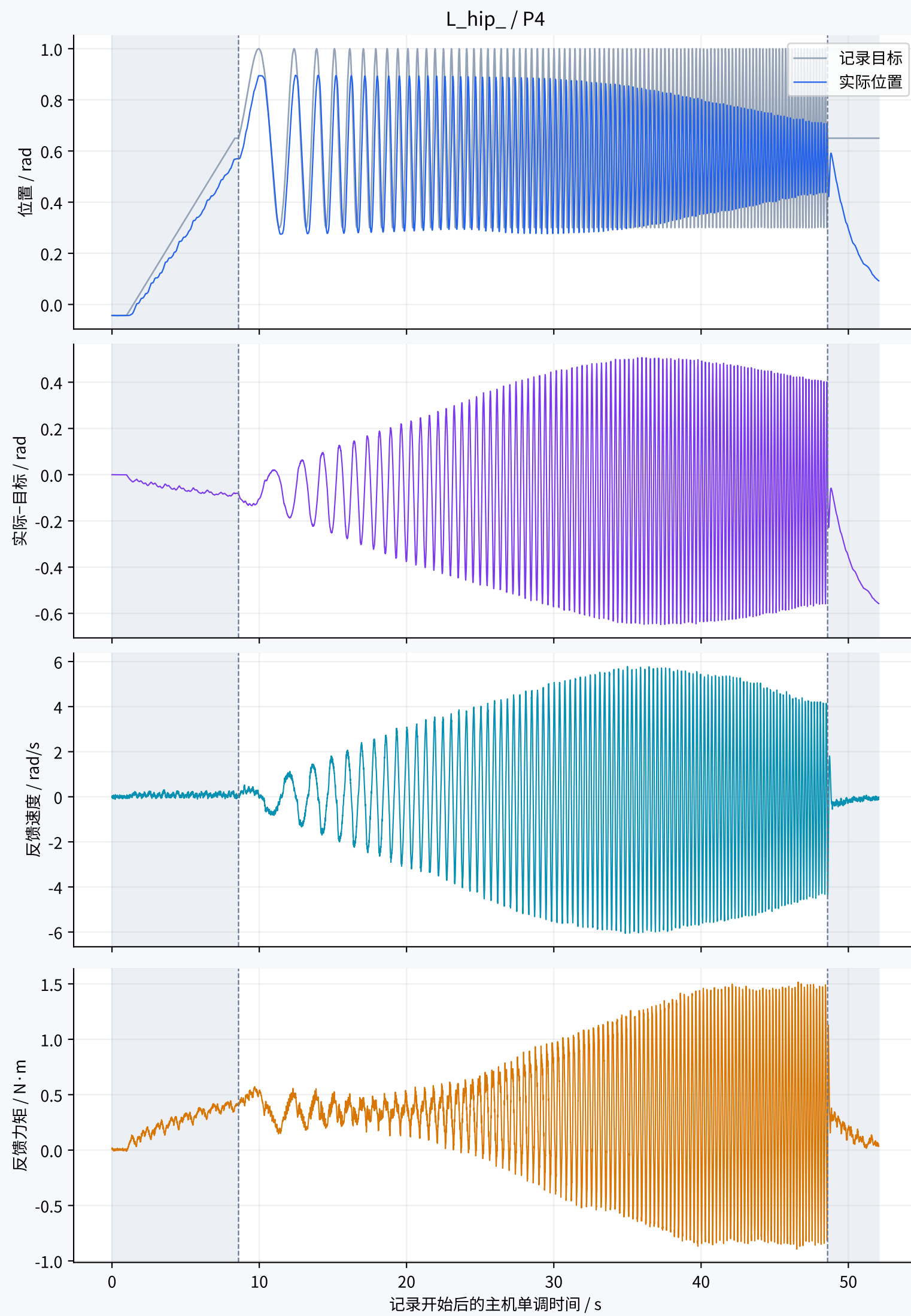
小腿 calf · 左侧



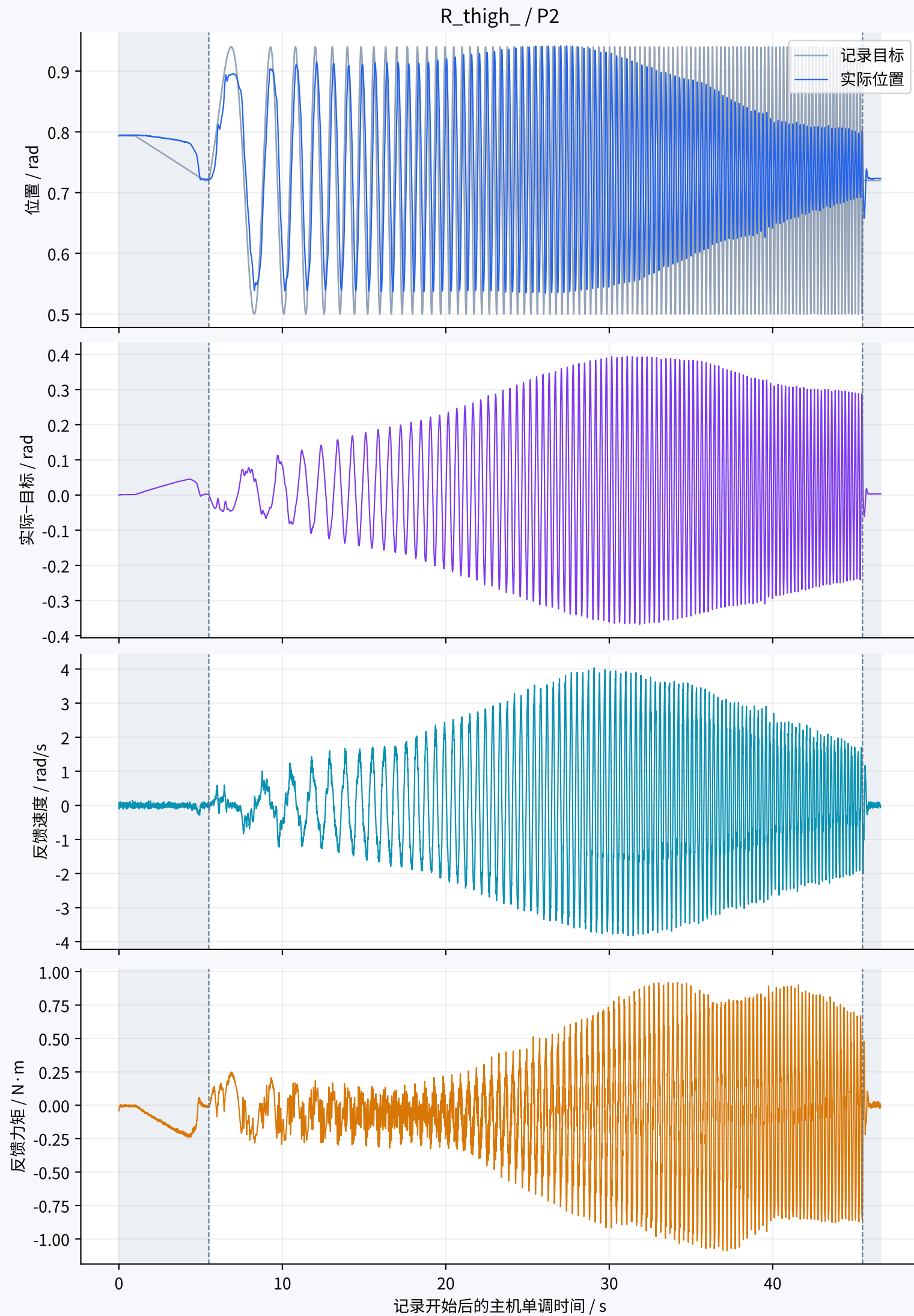
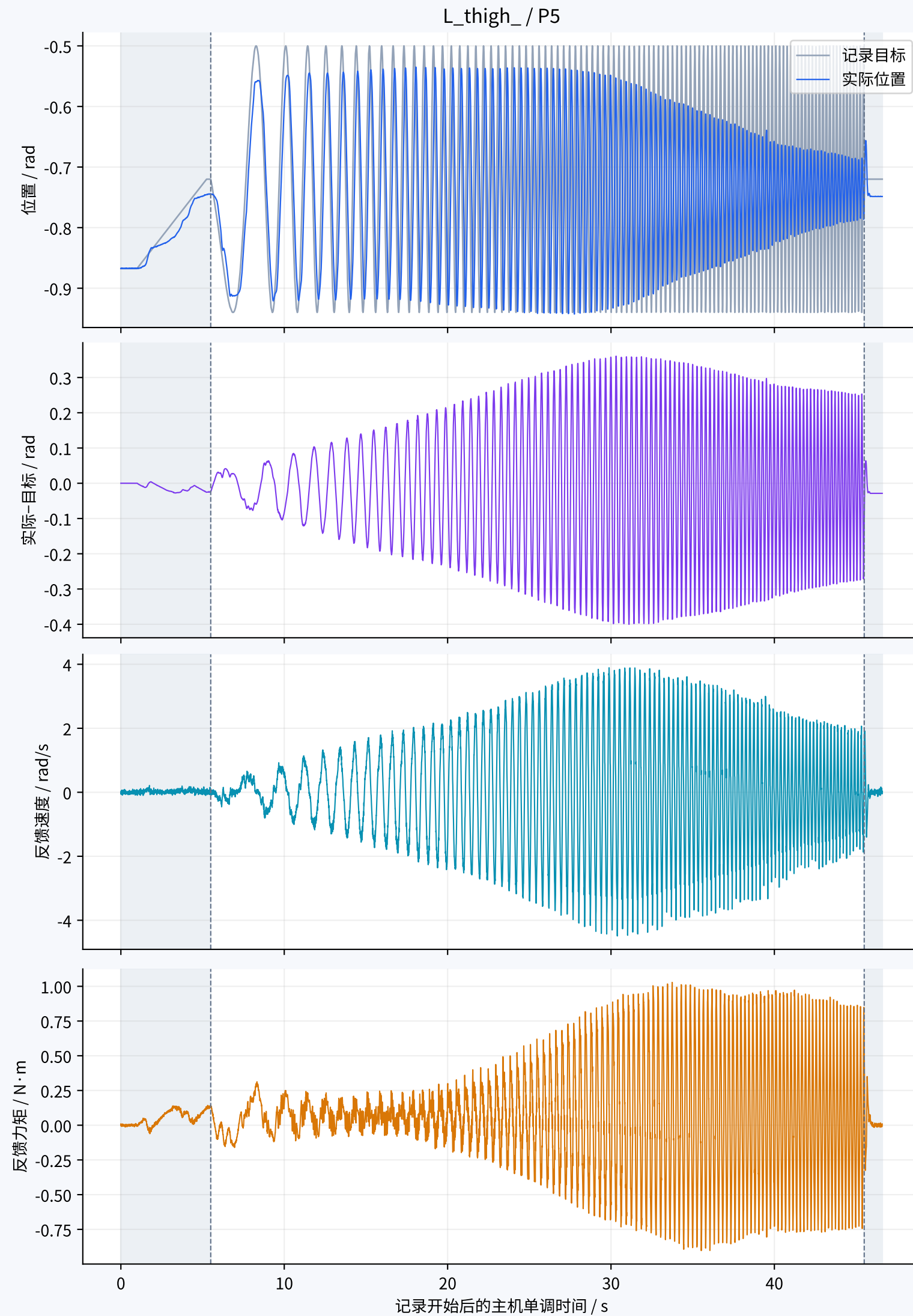
小腿 calf · 右侧



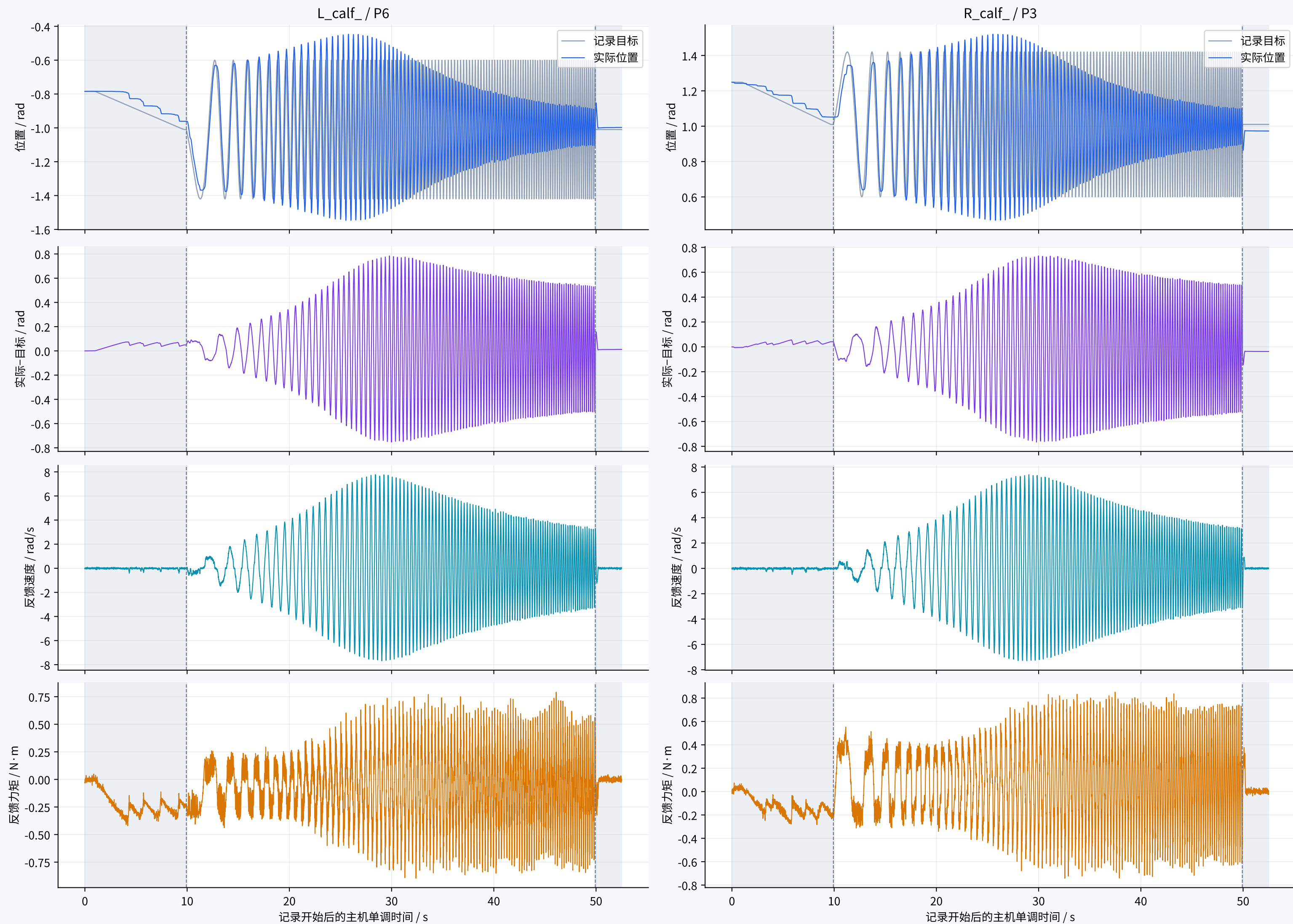
01\_A\_baseline\_hip | Kp=4.3, Kd=0.34  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



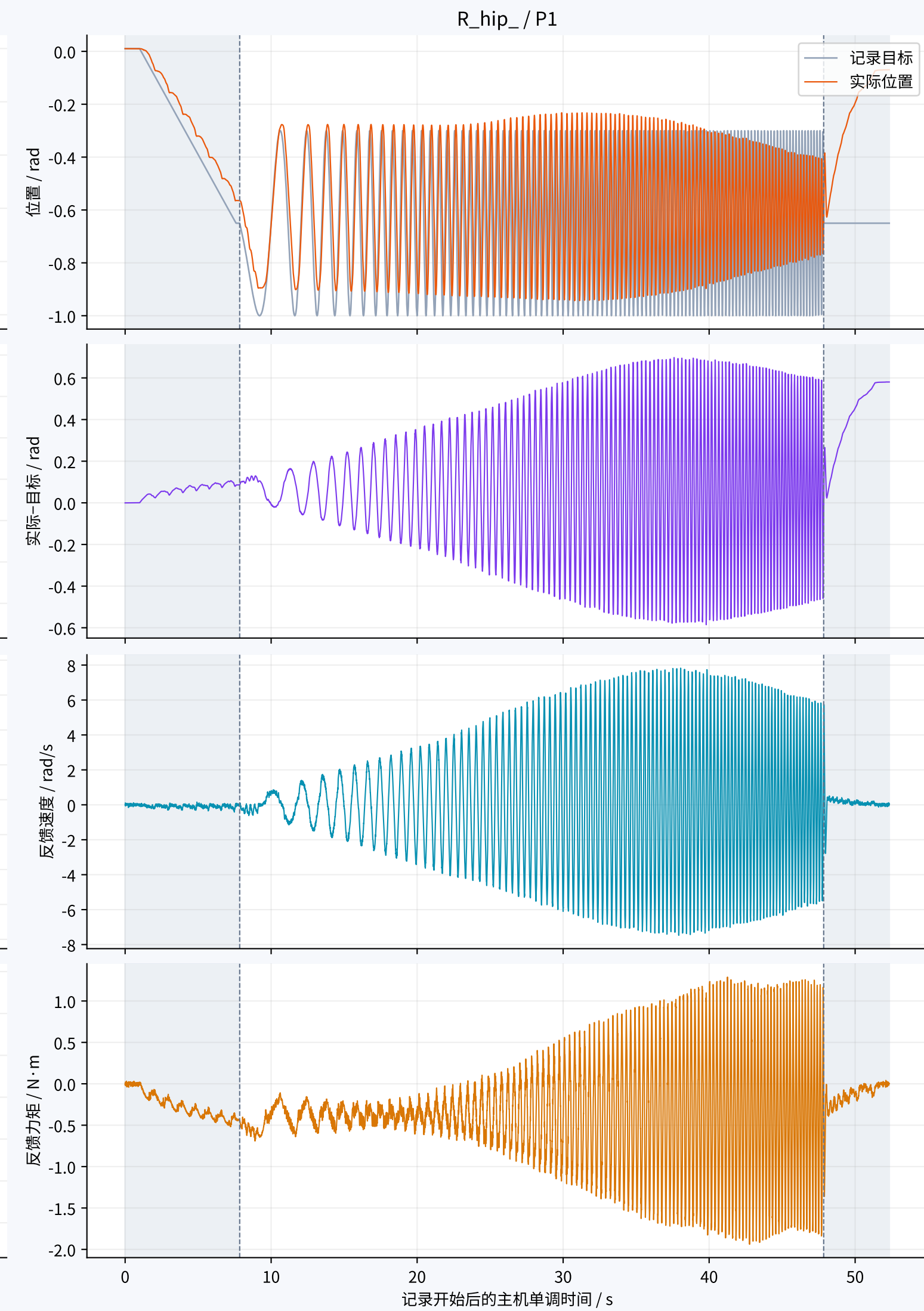
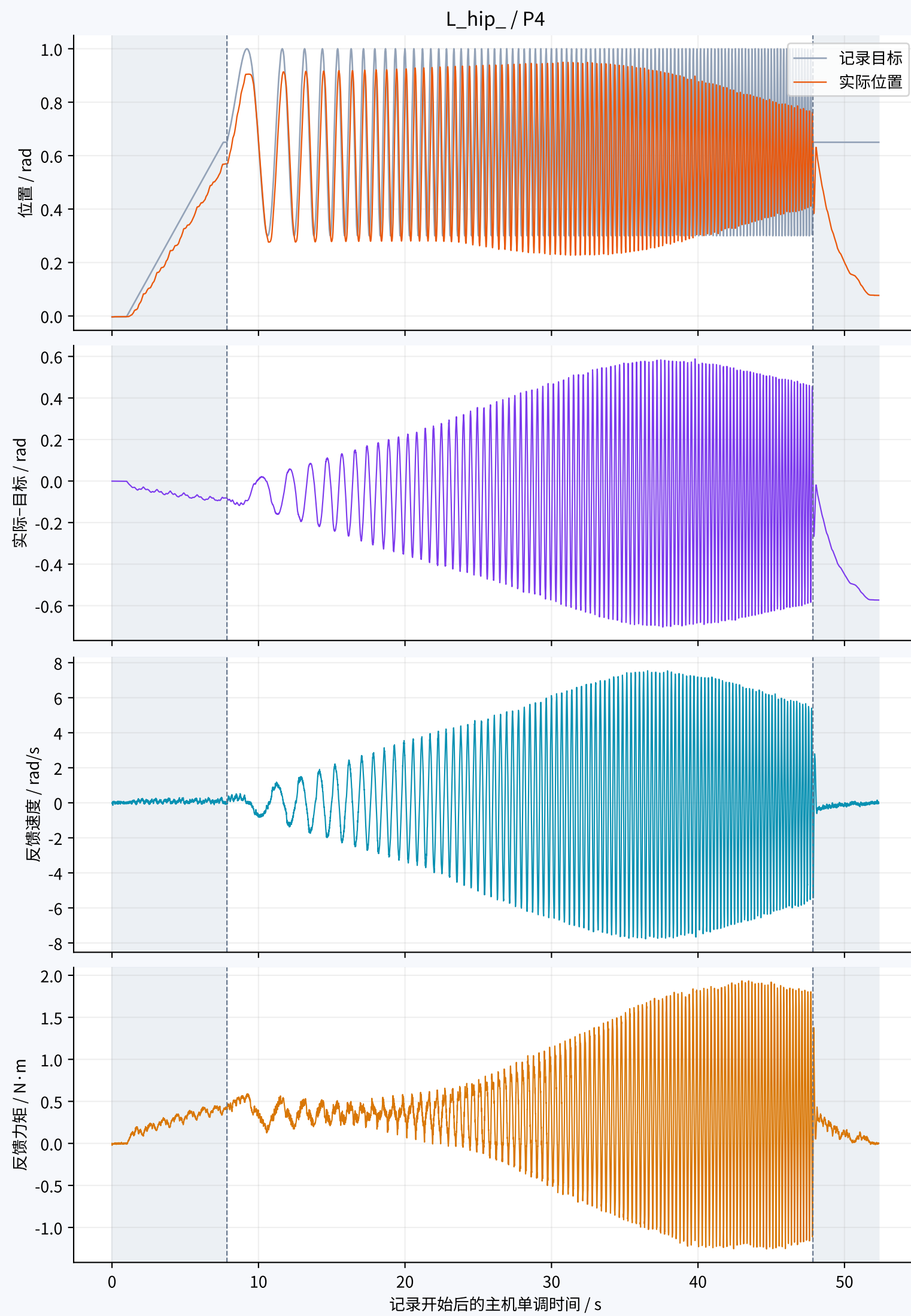
02\_A\_baseline\_thigh | Kp=4.3, Kd=0.34  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



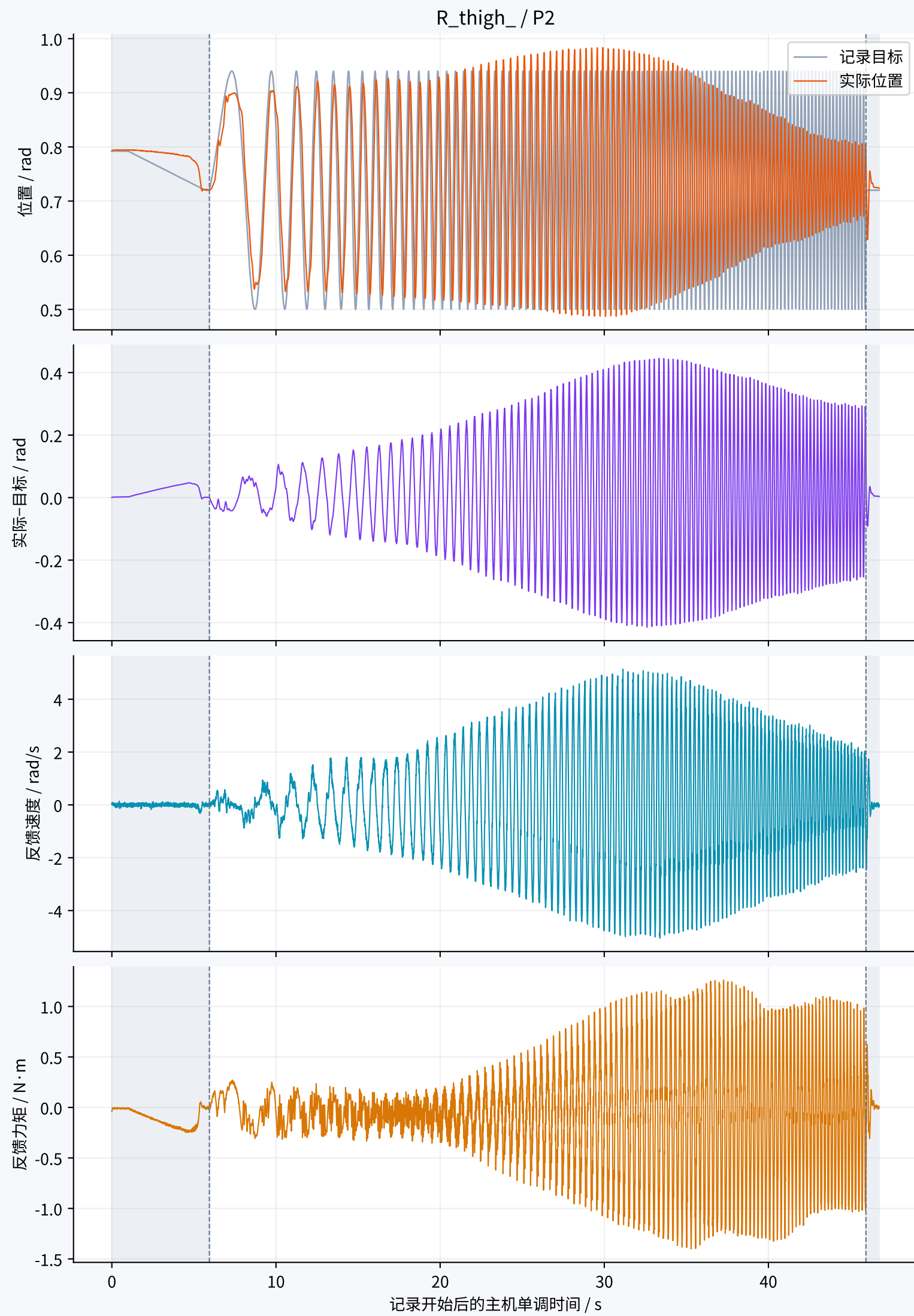
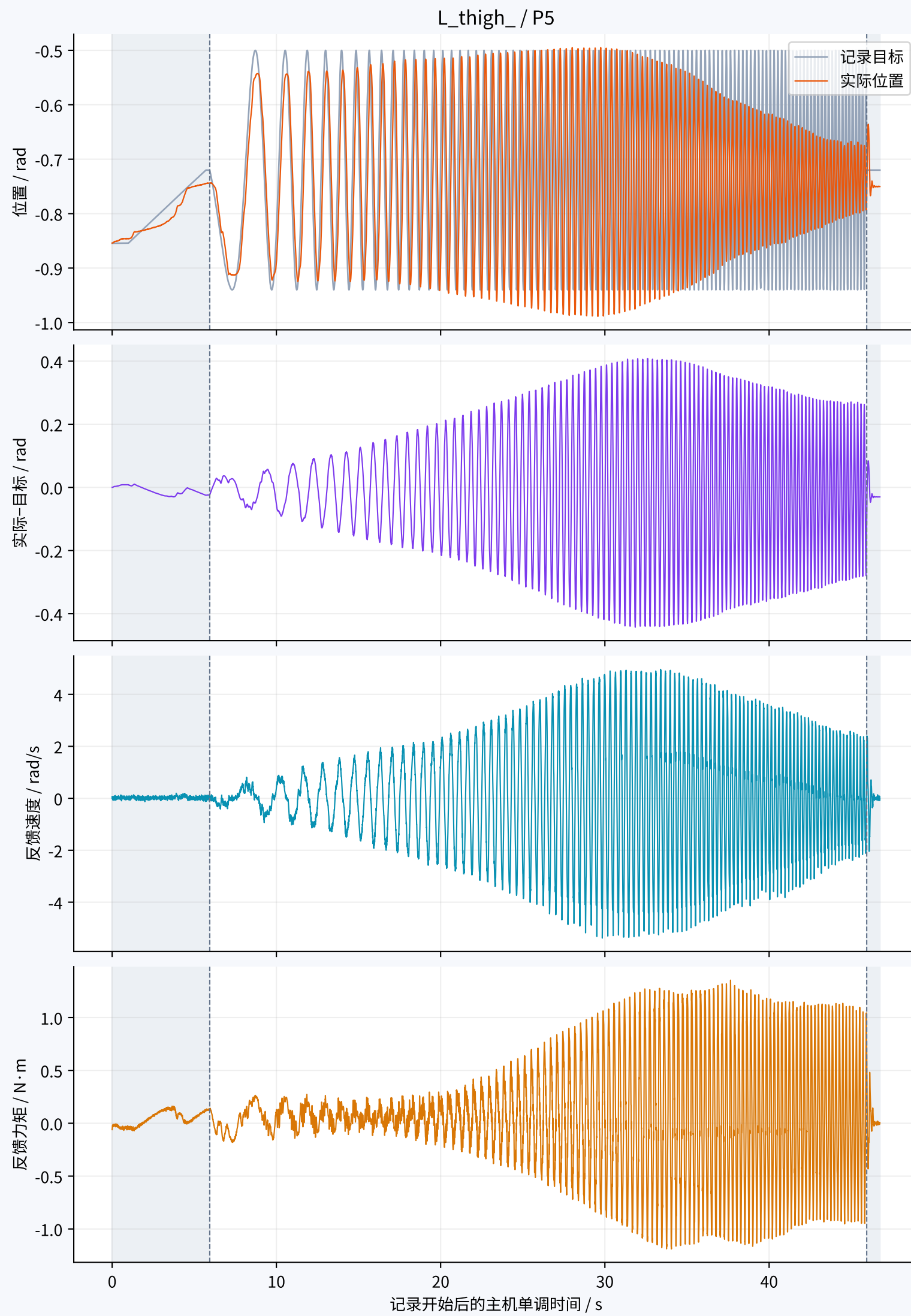
03\_A\_baseline\_calf | Kp=4.9, Kd=0.24  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



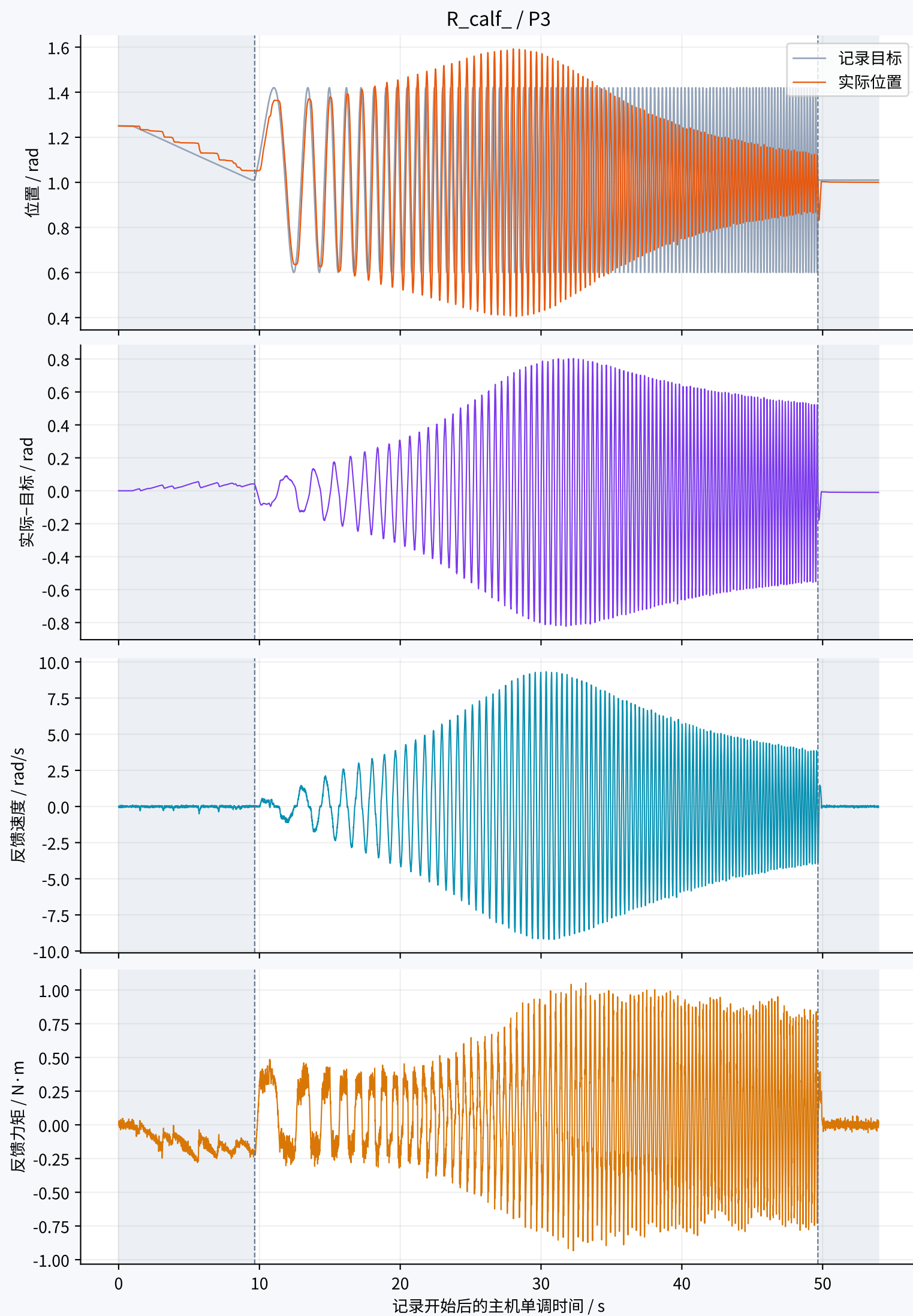
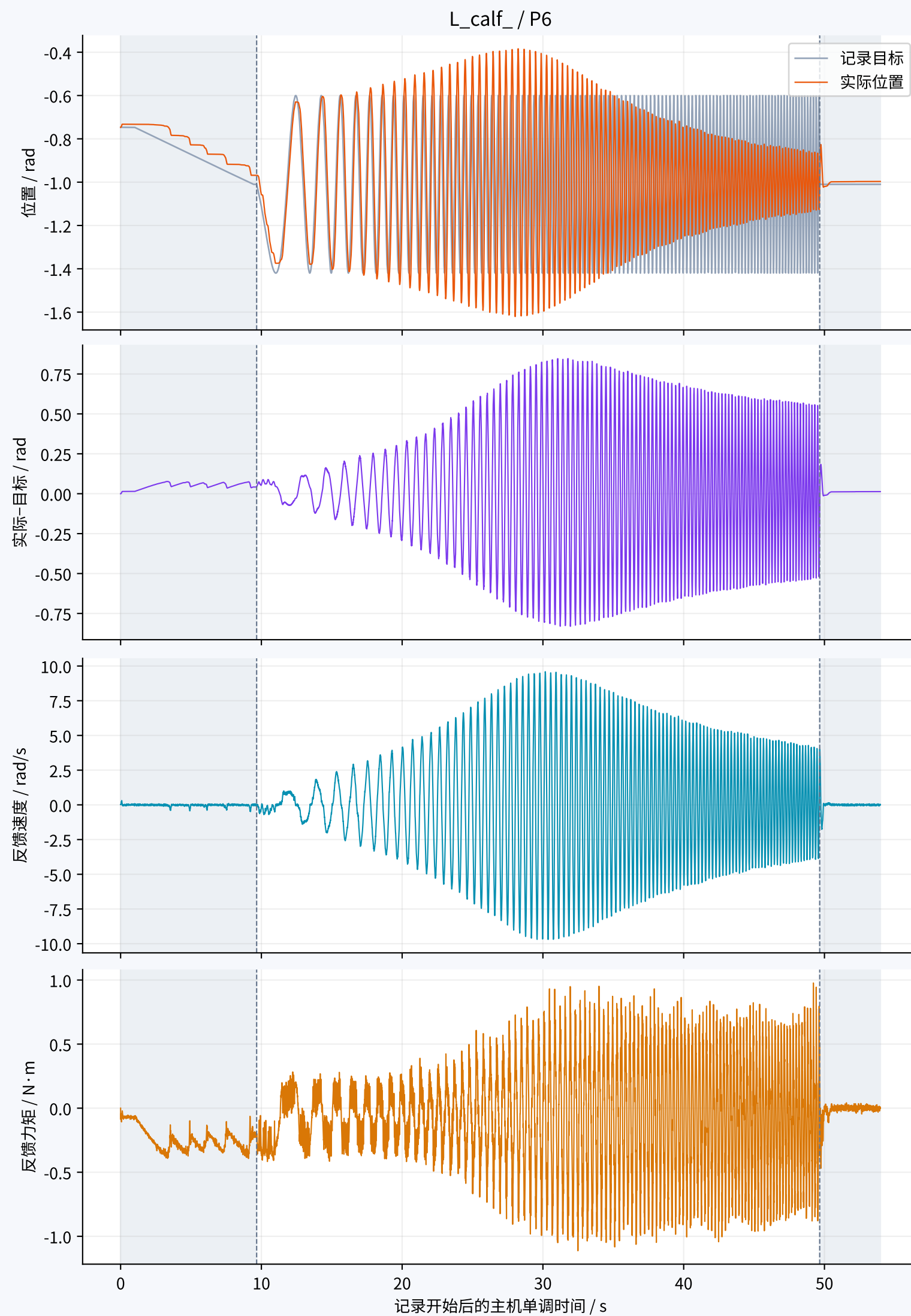
04\_B\_position\_hip | Kp=5.2, Kd=0.34  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



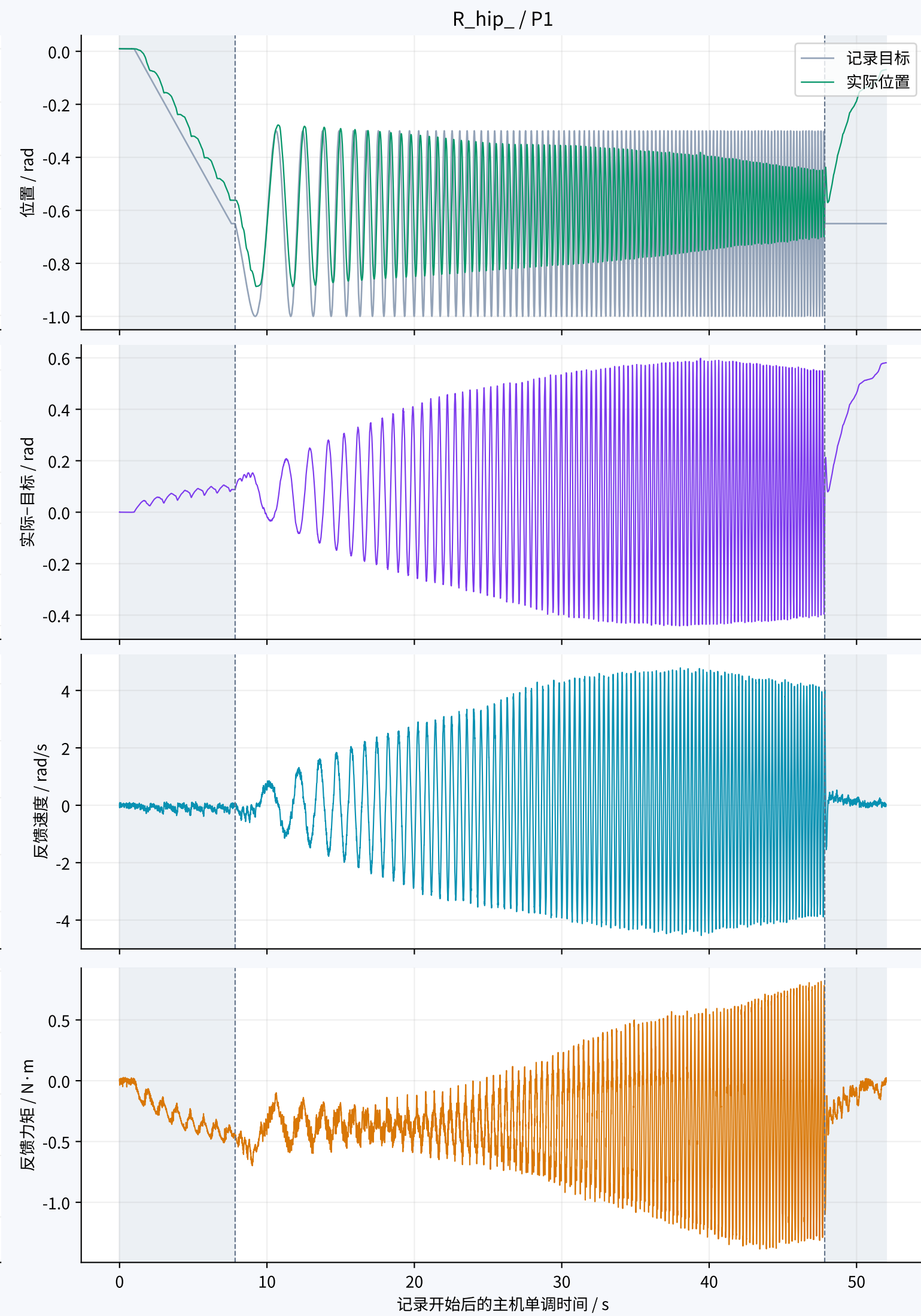
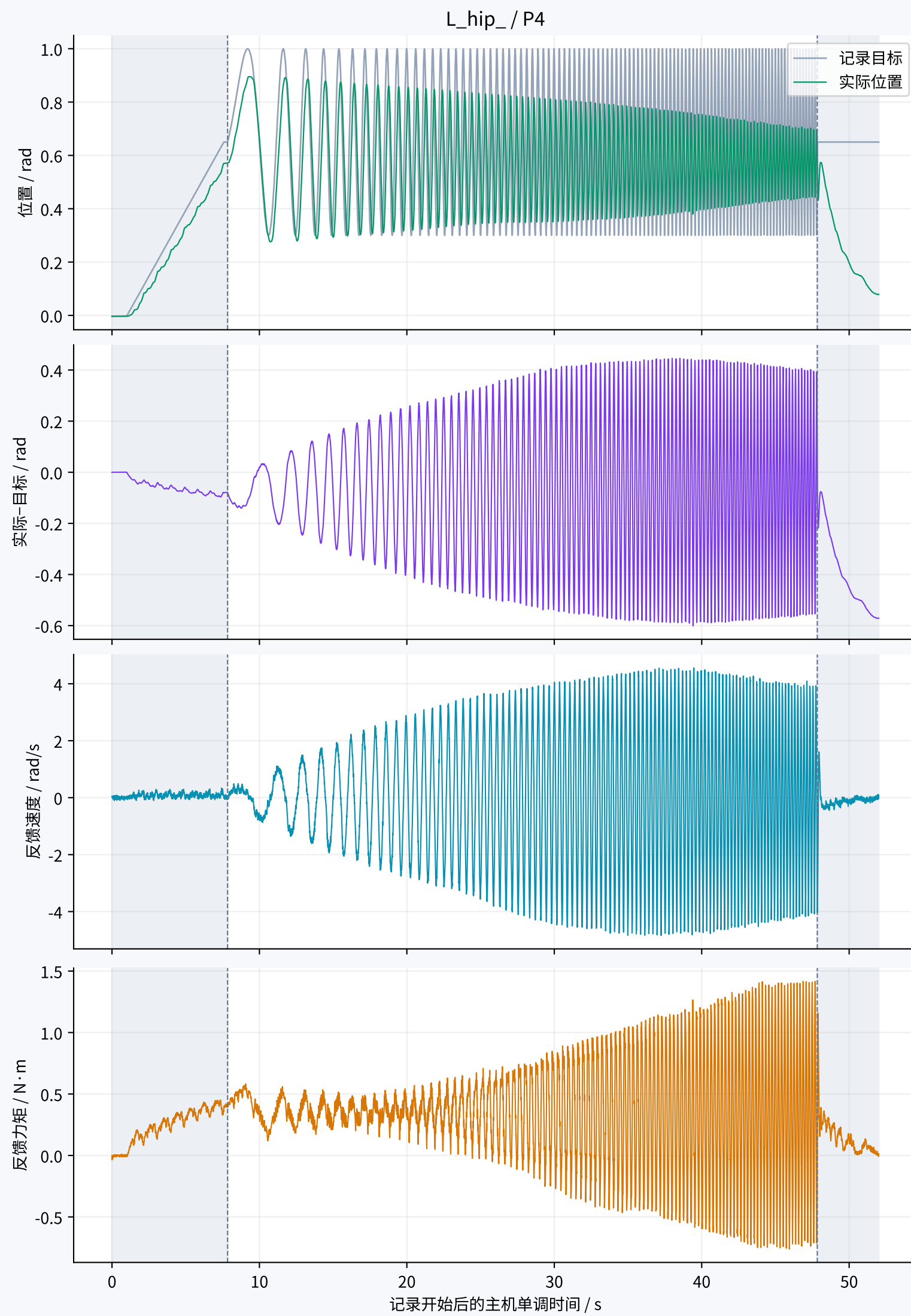
05\_B\_position\_thigh | Kp=5.2, Kd=0.34  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



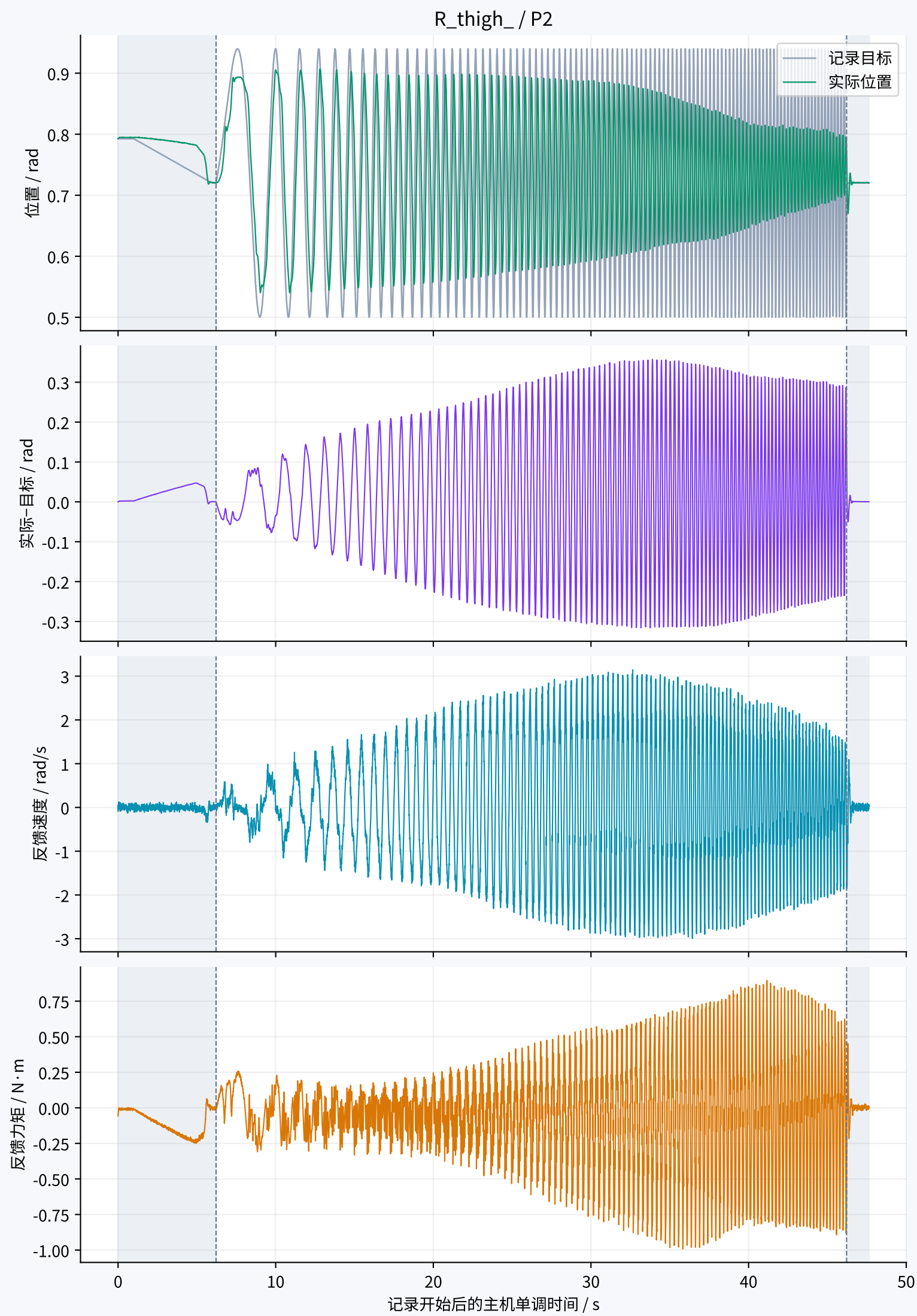
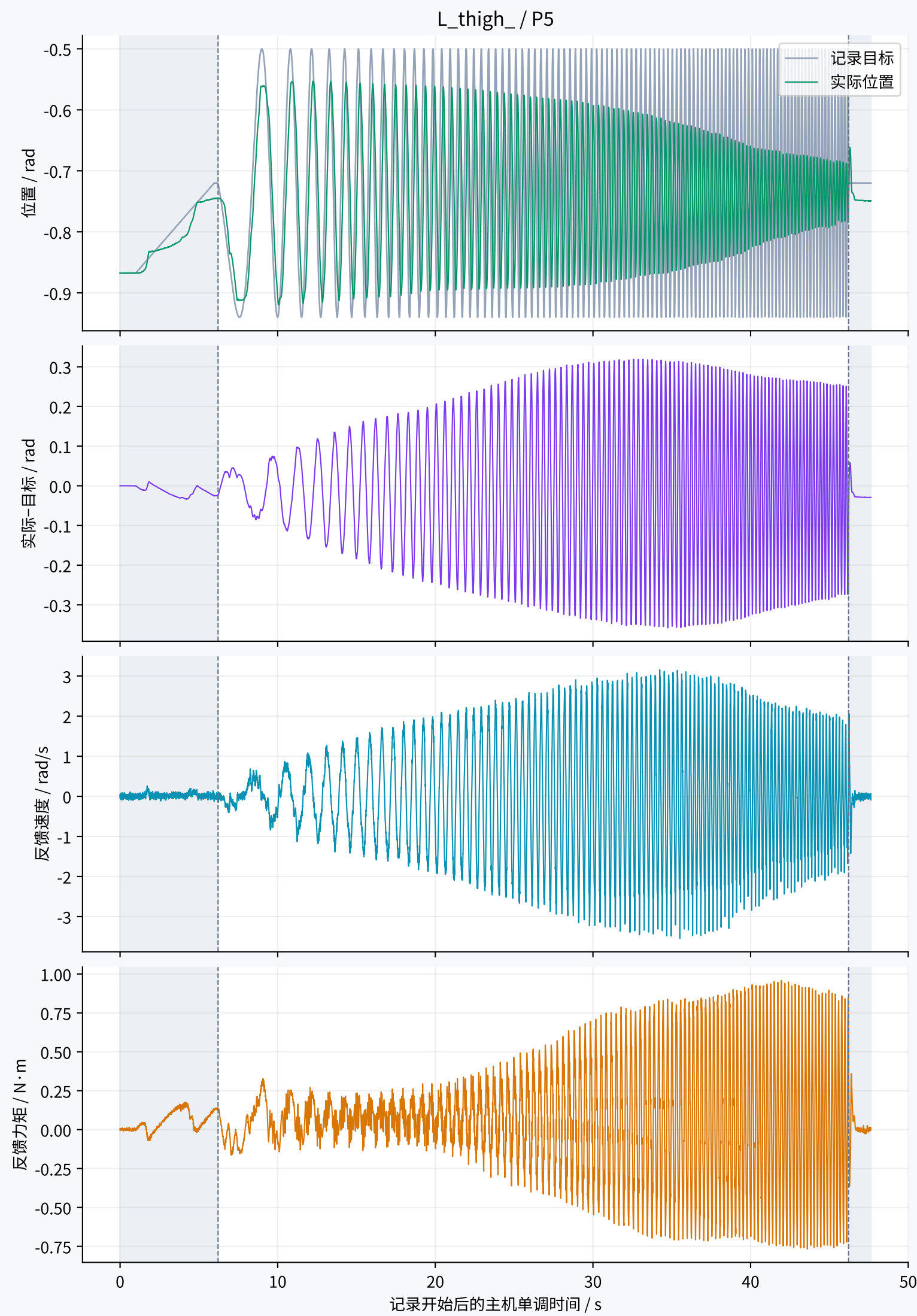
06\_B\_position\_calf | Kp=5.9, Kd=0.24  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



07\_C\_damping\_hip | Kp=4.3, Kd=0.42  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



08\_C\_damping\_thigh | Kp=4.3, Kd=0.42  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标



09\_C\_damping\_calf | Kp=4.9, Kd=0.3  
完整记录；灰色区域为正式扫频之外的准备/保持段；左右保留原关节坐标

